

A *loop* is a block of code that runs multiple times. Use loops when you need to repeat actions more than once, depending on certain criteria. The two loop types are the *for* loop and the *while* loop.

A *for* loop repeats as many times as you specify or once for each item in a sequence. For example, you can write a *for* loop that runs 100 times to repeat an action 100 times. You can also write a *for* loop that runs once for each item in a collection of items.

A *while* loop runs as long as a certain condition is true. Say you have a game loop that runs as long as a `game_active` variable is set to `True`. You can then put multiple situations inside the loop that cause `game_active` to become `False` and end the game. You can also use a *while* loop to process items in a collection that might have new items added to it as the loop is running. As long as items are in the collection, you can keep working with it. When the collection is empty, the loop stops running.

A *nested* loop is a loop inside another loop. Say you need a loop that examines every pixel in an image. You could write a loop that looks at each pixel in a row and, inside that loop, add a second loop that looks at every pixel in the current column for that row.